

Feature

Startup accelerators in India are doing a great job in identifying startups with potential.

**More Candy Crush**

Lionsgate TV and King Ltd are teaming up to produce a new TV game show for CBS based on Candy Crush series. <http://digit.in/CandyCrsh>



The future of design, material science and new ways of making things

In conversation with Tom Wujec from Autodesk

Tom Wujec,
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On the sidelines of the recent Autodesk University conference in Mumbai, we got a chance to speak to Tom Wujec, currently a Fellow at Autodesk and previously instrumental in creating Maya – one of leading 3D computer animation software in the industry today. In the freewheeling conversation below we talk about everything from the future of design, material science, digital humans and even the maker movement.

d If it's ok with you I'd like to keep our discussion free flowing. What would you like to start with?

Wujec: Since Digit is a technology magazine, why don't I talk about the impact of four particular technologies and how we see they're changing the way we design, make, and use just about everything – from running shoes to automom-

biles and even everyday items. Here's an amazing fact that I stumbled upon long ago: according to the american microprocessor association, we're now producing more individual transistors than grains of rice harvested on the planet and that's astonishing! We're now producing individual transistors on chips at the rate of 13 trillion per second. Essentially what that means is computation is becoming more available, distributed less expensive and more flexible than ever. There are three major classes of technologies that these chips are going into that are changing the way we design and make things. One is sensors, second is computation and third is fabrication. Sensors allow us to understand and read the world in ways in which we never could do with such precision. They come in all shapes and sizes and they can not only see better than people but also can hear, or taste, or smell, and perceive properties of the world that human beings cannot on their own.

d Right. In fact, there's a trend in which people are surgically augmenting themselves with these kind of sensors to detect magnetism, WiFi etc.

Wujec: Exactly! The second class of technology as we know, and your readers know, is the growth of computation. So Moore's Law is changing as a result of chips. But certainly the trend of increasing computation and calculating power is moving in parallel. What that means is we have increasingly more power to offload cognitive skills to the computer. Digital tools can do simulations, analysis, and increasingly machine learning, generative design and other classes of computation that were never previously available. And what that's doing is kind of allowing us to have magical powers in thinking and analysis.

The third tool is fabrication, which is essentially robotics. The same way sensors can read the world, robots can 'write' to the world – through actuators and manipulation. We have an explo-